

KOLEJ UNIVERSITI TUNKU ABDUL RAHMAN
FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY
ACADEMIC YEAR 2022/2023
OCTOBER EXAMINATION
AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE

MONDAY, 17 OCTOBER 2022

TIME: 9.00 AM – 11.00 AM (2 HOURS)

DIPLOMA IN COMPUTER SCIENCE

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

AACS3273 FUNDAMENTAL OF ARTIFICIAL INTELLIGENCE**Question 1**

- a) The *Turing Test*, originally known as imitation game, was introduced by Alan Turing in 1950. If a computer passes the test, it can be considered as intelligent. Explain **THREE (3)** capabilities a computer would need to pass the mentioned test. (6 marks)
- b) An intelligent agent is an entity that enables artificial intelligence to be put into action.
- (i) Explain what an intelligent agent is with an aid of a diagram. (1 + 4 marks)
- (ii) You are required to design an interactive math tutor, list the task environment and provide an example for each of the components. (4 + 4 marks)
- c) You are required to design an agent to solve the 8-puzzle problem as shown in **Figure 1**. Identify the formulation of the problem based on: (i) State; (ii) Initial state; (iii) Actions; (iv) Transition model; (v) Goal state; and (vi) Action cost.

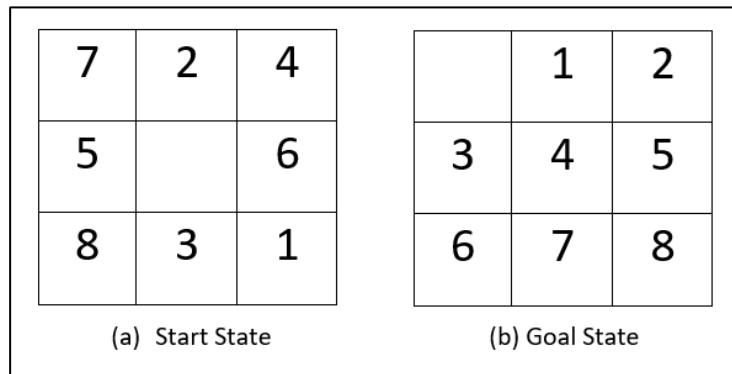


Figure 1: 8-Puzzle Problem

(6 marks)

[Total: 25 marks]

Question 2

- a) **Figure 2** shows some interconnected cities in Country X. Assume that you are building an agent to reach the destination, H from A.

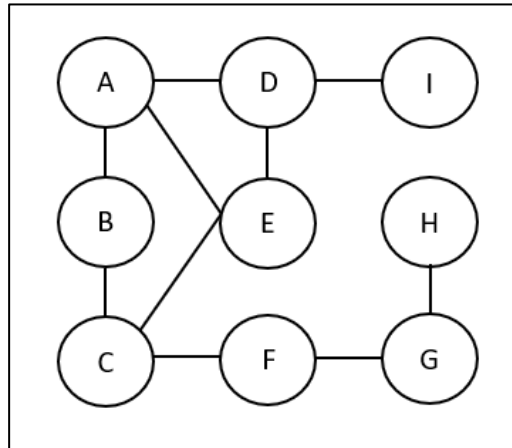


Figure 2: Interconnected cities in Country X

- (i) Illustrate the search tree, in ascending order, of a *Breadth-first search* (BFS) based on **Figure 2**. Stop when the destination is met. (4 marks)
 - (ii) Illustrate the search tree, in ascending order, of a *Depth-first search* (DFS) based on **Figure 2**. Stop when the destination is met. (4 marks)
 - (iii) Identify the solution path for BFS and DFS. (1 + 1 marks)
 - (iv) Depth-limited search is a variant of DFS that controls the depth of a search. However, it has its limitation. Explain the **TWO (2)** limitations. (4 marks)
- b) *Hill climbing search* is a local search algorithm that often gets stuck at a non-optimal solution. List and explain the **TWO (2)** problems in hill climbing. (6 marks)
- c) *Minimax* is an algorithm that is used in game theory to find the optimal move for a player. Compute the *Minimax* value for the nodes (A to E) in the game tree shown in **Figure 3**.

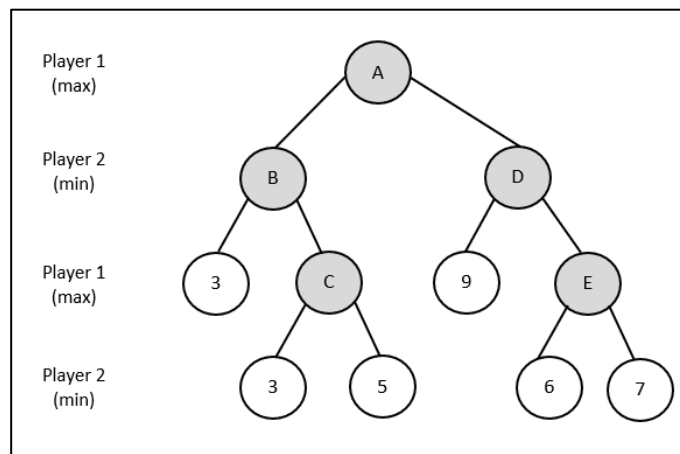
Question 2 c) (Continued)

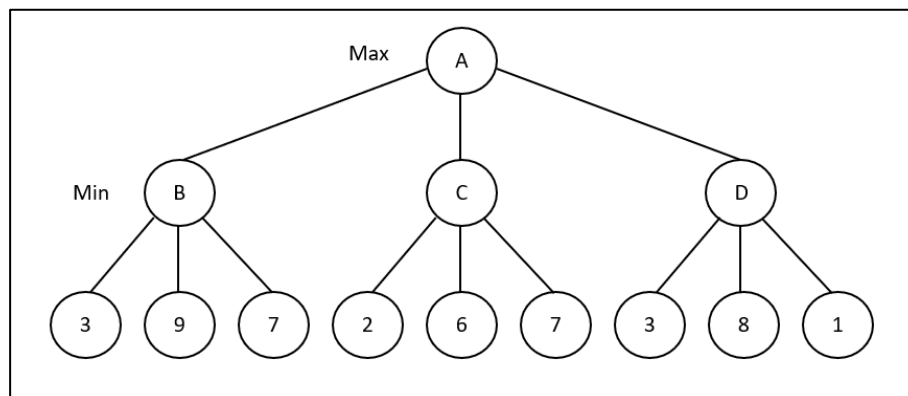
Figure 3: Game tree

(5 marks)

[Total: 25 marks]

Question 3

- a) *Alpha-beta pruning* is the pruning of useless branches to optimize the *Minimax* algorithm. **Figure 4** shows the search tree using the *Minimax* algorithm.

Figure 4: Search tree using the *Minimax* algorithm

- (i) Identify nodes or branches that would be pruned. Redraw the tree and indicate the pruned nodes or branches with crosses. (4 marks)
- (ii) Determine the Max's decision by highlighting the path or arcs in the tree together with the values of Node A to Node D. (6 marks)

AACS3273 FUNDAMENTAL OF ARTIFICIAL INTELLIGENCE

- b) Assuming you oversee schedule for the truck drivers in Company X. There are a total of 6 trips of services in the last week of December. However, only two drivers are available during the week due to the year end holiday season.

The hours are:

Trip 1: 9:00am - 2:00pm
 Trip 2: 9:30am - 3:00pm
 Trip 3: 3:00pm - 5:00pm
 Trip 4: 10:00pm – 3:00am
 Trip 5: 11:00pm – 2:00am
 Trip 6: 10:00am – 12:00pm

The drivers are:

Driver A, day shift (8am – 5pm).
 Driver B, night shift (10pm – 7am).

Formulate this problem as a *Constraint Satisfaction Problems (CSP)* and identify:

- (i) The variables and the domains for each of the variable. (6 marks)
- (ii) The constraints. (4 marks)
- (iii) Construct the constraint graph according to your CSP. (5 marks)

[Total: 25 marks]

Question 4

- a) Knowledge-based agents make use of reasoning processes that operate on some internal representations of knowledge. The central component of a knowledge-based agent is its knowledge base.

- (i) Define knowledge base. (2 marks)
- (ii) List and explain **TWO (2)** main operations involved a knowledge base. (4 marks)

- b) You are hiding an egg from your brothers and the clues of the location of the egg are as shown in **Question 4 b) (i) – (v)**. Let:

- p = “The house is next to a lake”
- q = “The egg is in the kitchen”
- h = “The plant in front of the house is a hibiscus”
- r = “The plant in the back yard is a rose”
- g = “The egg is in the garage”
- f = “The egg is in the freezer”

Translate the sentences below into proposition formulas with the correct logical connectives.

- (i) If the house is next to a lake, then the egg is in the kitchen. (1 mark)
- (ii) If the house is not next to a lake or the egg is in the freezer, then the plant in front of the house is a hibiscus and the plant in the back yard is not a rose. (1 mark)

AACS3273 FUNDAMENTAL OF ARTIFICIAL INTELLIGENCE**Question 4 b) (Continued)**

- (iii) If the egg is in the garage, then the plant in the back yard is not a rose. (1 mark)
- (iv) If the egg is not in the freezer, then the plant in front of the house is not a hibiscus. (1 mark)
- (v) The egg is not in the kitchen. (1 mark)
- c) Construct a truth table for each of the following propositions and identify if it is a tautology.
- (i) $(p \vee q) \vee \neg p \wedge \neg q$ (6 marks)
- (ii) $p \vee q \Rightarrow \neg(p \wedge q) \wedge \neg p$ (6 marks)
- d) Explain the difference between *proposition logic* and *first-order logic*. (2 marks)

[Total: 25 marks]